Procrustes and RANSAC

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Announcement

- HW1 is out, due Feb 18, 11:59 pm
- Written + coding on GradeScope
- Autograder for coding part (100 points)
- Bonus points

Recap: pose from homography (planar)

Normalize

$$\widehat{\mathbf{H}} = \mathbf{K}^{-1}\mathbf{H} = \left[\, \hat{\boldsymbol{h}}_1 \; \hat{\boldsymbol{h}}_2 \; \hat{\boldsymbol{h}}_3 \,\right].$$

Recover pose up to a scale

Choose scale s (one common choice):

$$s = \frac{1}{\|\hat{\boldsymbol{h}}_1\|} \quad (\approx \frac{1}{\|\hat{\boldsymbol{h}}_2\|}).$$

Then

$$\mathbf{R}_1 = s\,\hat{\boldsymbol{h}}_1, \quad \mathbf{R}_2 = s\,\hat{\boldsymbol{h}}_2, \quad \mathbf{t} = s\,\hat{\boldsymbol{h}}_3, \quad \mathbf{R}_3 = \mathbf{R}_1 \times \mathbf{R}_2.$$

Recap: pose from homography

Because $\mathbf{H}$ is estimated from noisy data, the recovered columns are not perfectly orthonormal.

Project to the closest rotation

Form $\widetilde{\mathbf{R}} = [\mathbf{R}_1\ \mathbf{R}_2\ \mathbf{R}_3]$ and compute SVD:

$$\mathbf{R} = \mathbf{U}\,\mathrm{diag}(1, 1, \det(\mathbf{U}\mathbf{V}^\top))\,\mathbf{V}^\top.$$

Cheirality (positive depth)

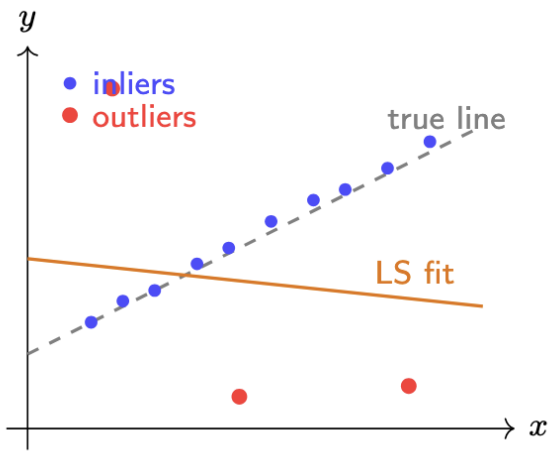
Pick the sign/scale so that the tag's known 3D points (on the plane $Z = 0$) transformed by the candidate pose

$$\boldsymbol{X}_c = \mathbf{R}\boldsymbol{X} + \mathbf{t}$$

Recap: PnP

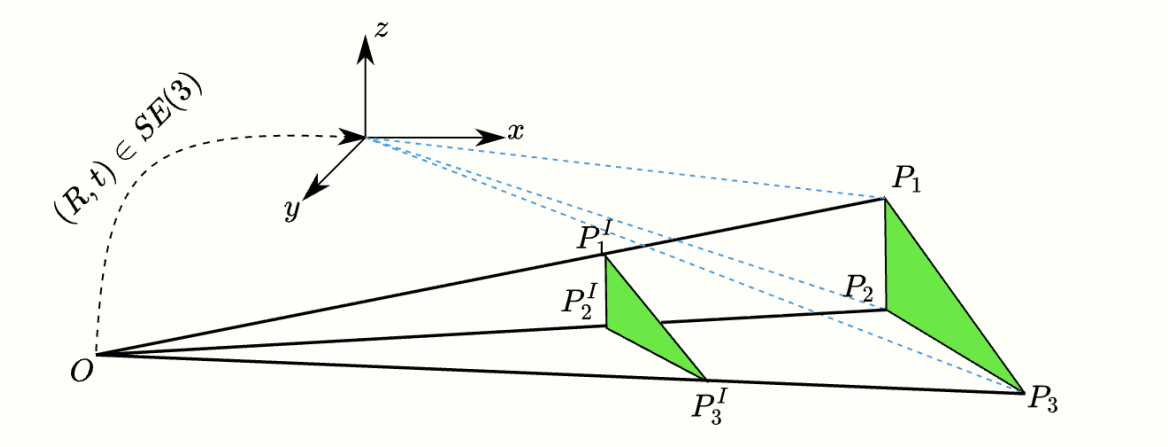
Non-linear Least Squares of re-projection error

- Outliers introduce errors
- Bad initialization prevents converger



Recap: P3P

Bearing problem



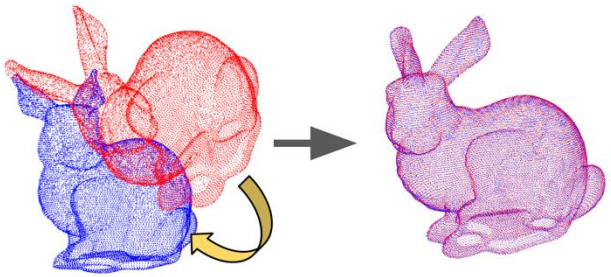
1. **Pixels → rays:** $\boldsymbol{u}_i \propto \mathbf{K}^{-1}[u_i \ v_i \ 1]^\top$, normalize $\|\boldsymbol{u}_i\| = 1$.
2. **Solve depths:** use cosine-law constraints with $d_{ij} = \|\boldsymbol{P}_i - \boldsymbol{P}_j\|$ and $c_{ij} = \boldsymbol{u}_i^\top \boldsymbol{u}_j$ to get (d_1, d_2, d_3).
3. **Lift to 3D:** $\boldsymbol{p}_i = d_i \boldsymbol{u}_i$.
4. **Pose via Procrustes:** find $(\mathbf{R}, \mathbf{t})$ s.t. $\boldsymbol{p}_i \approx \mathbf{R}\boldsymbol{P}_i + \mathbf{t}$. Our topic today
5. **Disambiguate:** choose candidate $(\mathbf{R}^{(k)}, \mathbf{t}^{(k)})$ with cheirality + smallest reprojection error (use $(\boldsymbol{P}_4, \boldsymbol{x}_4)$ if available).

Question: homography vs P3P

Why does homography need 4 points, but P3P only needs 3 points?

- Homography has 8 degrees of freedom (not only pose but also point mapping)
- P3P assumes a calibrated camera, and the full pose has 6 DoF

Procrustes problem (rigid)

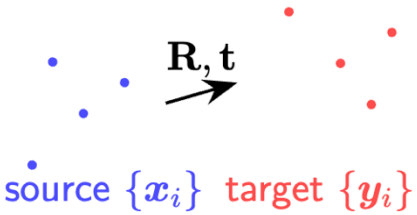


Given N correspondences $\{(\boldsymbol{x}_i, \boldsymbol{y}_i)\}_{i=1}^N$ with $\boldsymbol{x}_i, \boldsymbol{y}_i \in \mathbb{R}^d$.

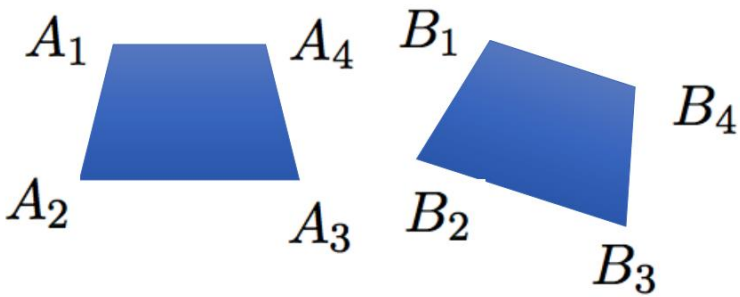
Rigid Procrustes objective

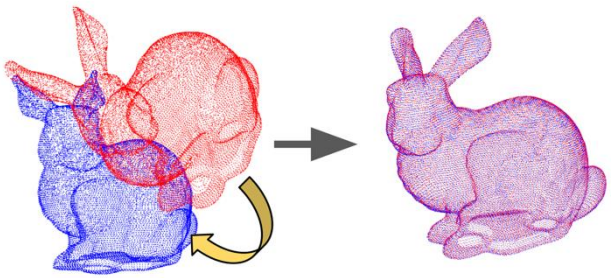
$$\min_{\mathbf{R}\in \mathrm{SO}(d),\ \mathbf{t}\in\mathbb{R}^d} \sum_{i=1}^{N} \|\mathbf{R}\boldsymbol{x}_i + \mathbf{t} - \boldsymbol{y}_i\|_2^2.$$

- $\mathbf{R}$ is a rotation (orthonormal, $\det(\mathbf{R}) = +1$)
- $\mathbf{t}$ is a translation



Known correspondence





How do we solve for R, T from n point correspondences?

$$A_i = RB_i + T$$

If we don't have correspondence, it becomes a harder "registration" problem.

Step 1: centering

$$f(\mathbf{t}) = \sum_{i=1}^{N} \|\mathbf{R}\boldsymbol{x}_i + \mathbf{t} - \boldsymbol{y}_i\|_2^2.$$

Differentiate and set to zero:

$$\frac{\partial f}{\partial \mathbf{t}} = 2\sum_{i=1}^{N}(\mathbf{R}\boldsymbol{x}_i + \mathbf{t} - \boldsymbol{y}_i) = 0.$$

So

$$N\mathbf{t} = \sum_{i=1}^{N} \boldsymbol{y}_i - \mathbf{R}\sum_{i=1}^{N} \boldsymbol{x}_i \quad \Longrightarrow \quad \boxed{\mathbf{t}^* = \bar{\boldsymbol{y}} - \mathbf{R}\,\bar{\boldsymbol{x}},}$$

where $\bar{\boldsymbol{x}} = \frac{1}{N}\sum_i \boldsymbol{x}_i$ and $\bar{\boldsymbol{y}} = \frac{1}{N}\sum_i \boldsymbol{y}_i$.

The best estimate of t given R

We can get translation after solving R

Step 2: substituting t back

Define centered points:

$$\boldsymbol{x}_i' = \boldsymbol{x}_i - \bar{\boldsymbol{x}}, \qquad \boldsymbol{y}_i' = \boldsymbol{y}_i - \bar{\boldsymbol{y}}.$$

$$\min_{\mathbf{R} \in \mathrm{SO}(d)} \|\mathbf{R}\mathbf{X}_c - \mathbf{Y}_c\|_F^2.$$

Expand the centered objective:

$$\|\mathbf{R}\mathbf{X}_c - \mathbf{Y}_c\|_F^2 = \boxed{\|\mathbf{R}\mathbf{X}_c\|_F^2} + \|\mathbf{Y}_c\|_F^2 - \boxed{2\,\mathrm{tr}(\mathbf{R}\mathbf{X}_c\mathbf{Y}_c^\top).}$$

Maximizing the trace

Because $\mathbf{R}$ is orthonormal, $\|\mathbf{R}\mathbf{X}_c\|_F^2 = \|\mathbf{X}_c\|_F^2$ (constant).

Kabsch algorithm

$$\max_{\mathbf{R}\in \mathrm{SO}(d)} \ \mathrm{tr}(\mathbf{R}\mathbf{H}), \qquad \mathbf{H} = \mathbf{X}_c \mathbf{Y}_c^\top \in \mathbb{R}^{d\times d}.$$

Take an SVD:

$$\mathbf{H} = \mathbf{U}\,\mathbf{\Sigma}\,\mathbf{V}^\top.$$

Candidate rotation

$$\mathbf{R}_0 = \mathbf{V}\mathbf{U}^\top.$$

Sometimes $\mathbf{R}_0$ is a reflection (determinant -1).

Define

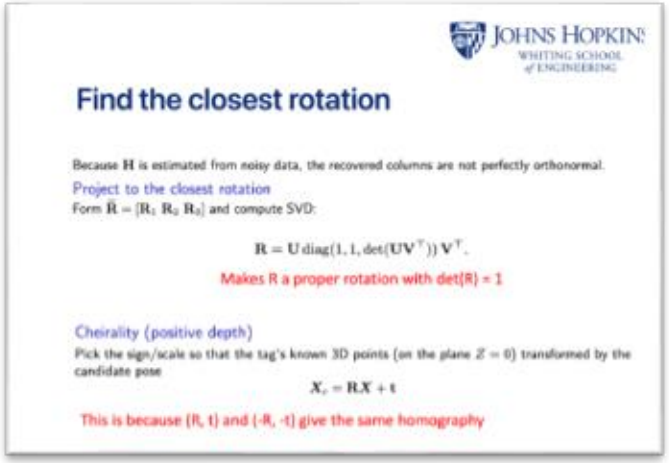
$$\mathbf{D} = \mathrm{diag}(1, \ldots, 1, \det(\mathbf{V}\mathbf{U}^\top)).$$

Then

$$\mathbf{R}^* = \mathbf{V}\,\mathbf{D}\,\mathbf{U}^\top \in \mathrm{SO}(d).$$

This is the standard Kabsch "sign correction".

Familiar?



Frobenius norm

For a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, the Frobenius norm is

$$\|\mathbf{A}\|_F = \sqrt{\sum_{i=1}^{m} \sum_{j=1}^{n} A_{ij}^2} = \sqrt{\mathrm{tr}(\mathbf{A}^\top \mathbf{A})} = \|\mathrm{vec}(\mathbf{A})\|_2.$$

Frobenius norm doesn't change after pre-multiplying by an orthonormal vector

Let $\mathbf{R} \in \mathbb{R}^{m \times m}$ be a rotation/orthogonal matrix, so $\mathbf{R}^\top \mathbf{R} = \mathbf{I}$.

Compute using the trace form

$$\|\mathbf{RA}\|_F^2 = \mathrm{tr}\big((\mathbf{RA})^\top (\mathbf{RA})\big) = \mathrm{tr}\big(\mathbf{A}^\top \mathbf{R}^\top \mathbf{RA}\big) = \mathrm{tr}(\mathbf{A}^\top \mathbf{A}) = \|\mathbf{A}\|_F^2.$$

Summary

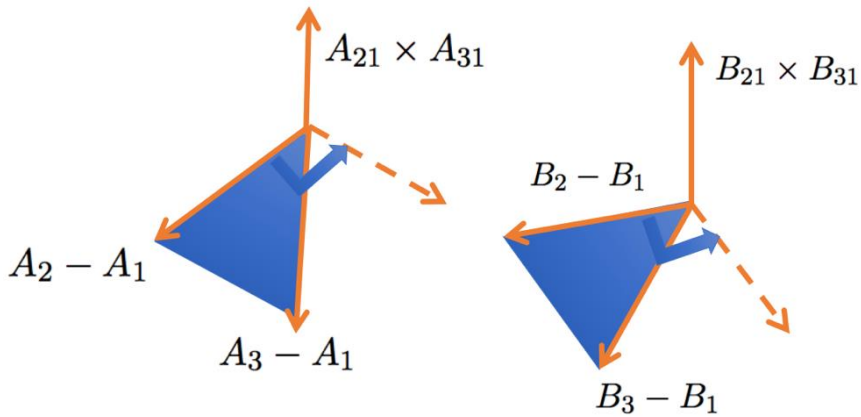
Input

Correspondences $\{(\boldsymbol{x}_i, \boldsymbol{y}_i)\}$.

Steps

1. Compute means $\bar{\boldsymbol{x}}, \bar{\boldsymbol{y}}$ and centered matrices $\mathbf{X}_c, \mathbf{Y}_c$.
2. Compute $\mathbf{H} = \mathbf{X}_c \mathbf{Y}_c^\top$.
3. SVD: $\mathbf{H} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$.
4. Rotation: $\mathbf{R}^* = \mathbf{V}\mathbf{D}\mathbf{U}^\top$ (with reflection fix).
5. Translation: $\mathbf{t}^* = \bar{\boldsymbol{y}} - \mathbf{R}^* \bar{\boldsymbol{x}}$.

Minimum number of points needed



Three non-collinear points suffice: each triangle $A_{i=1\ldots3}$ and $B_{i=1\ldots3}$ make an orthogonal basis

$$\begin{pmatrix} A_{21} & (A_{21} \times A_{31}) \times A_{21} & A_{21} \times A_{31} \end{pmatrix}$$

and

$$\begin{pmatrix} B_{21} & (B_{21} \times B_{31}) \times B_{21} & B_{21} \times B_{31} \end{pmatrix}$$

Rotation between two orthogonal bases is unique.

Outlier Rejection
RANSAC and Hough Transform

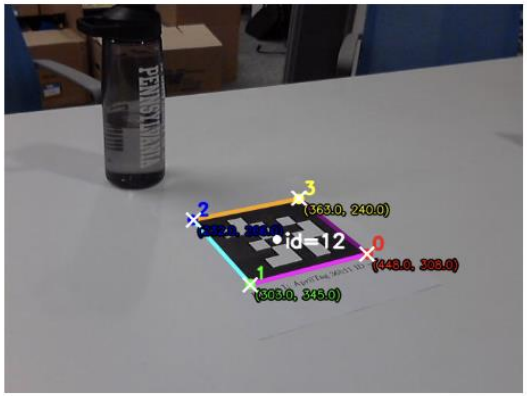
From last lecture..

What the pose solver wants

- Clean corner locations
- Or at least clean lines/edges that define corners
- A small number of gross outliers

Reality

- Specularities, motion blur, partial occlusion
- Spurious edges and bad correspondences
- Thresholding and corner localization produce outliers



Two classic tools for outlier rejection

RANSAC (robust fitting)

- You have a parametric model (e.g., a line)
- Many measurements are wrong (outliers)
- Find a model supported by many inliers

Hough (voting / detection)

- You have many local measurements (edge pixels)
- Each pixel votes for all models passing through it
- Peaks in an accumulator reveal strong global structure

RANSAC (Random Sample Consensus)

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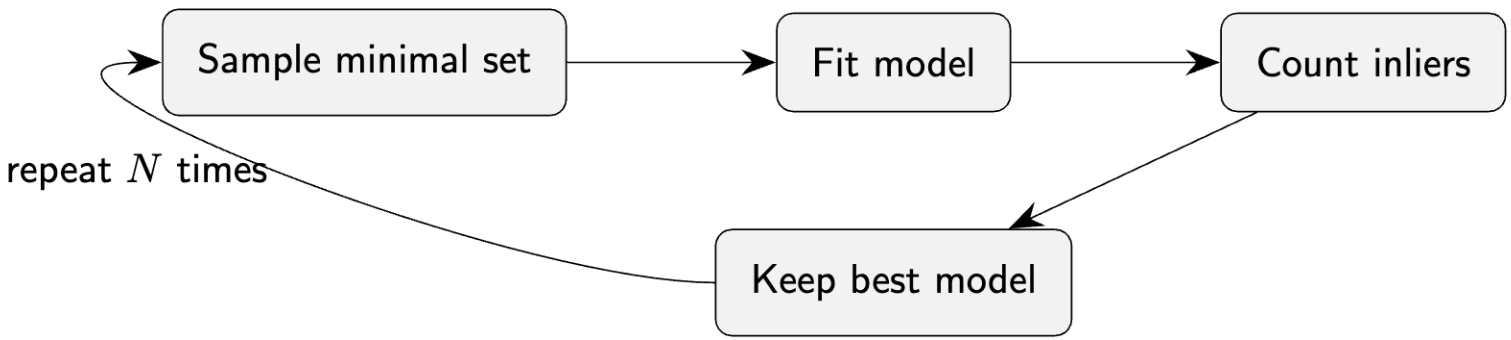


RANSAC = random hypotheses + a hard inlier test.

RANSAC – Sample minimal set

RANSAC needs a minimal number of points to instantiate a model.

Examples

- Line in 2D: $s = 2$ points
- Circle in 2D: $s = 3$ points
- Homography: $s = 4$ point correspondences (planar)

Number of points <-> DoF of objects

RANSAC – Fit model

Given points $\{(x_i, y_i)\}$, fit a 2D line.

$$\tilde{\boldsymbol{p}}_1 = \begin{bmatrix} x_1 \\ y_1 \\ 1 \end{bmatrix}, \qquad \tilde{\boldsymbol{p}}_2 = \begin{bmatrix} x_2 \\ y_2 \\ 1 \end{bmatrix}.$$

Estimate a line

The homogeneous line is the cross product:

$$\ell = \tilde{\boldsymbol{p}}_1 \times \tilde{\boldsymbol{p}}_2 = \begin{bmatrix} a \\ b \\ c \end{bmatrix}.$$

Distance between a point to line

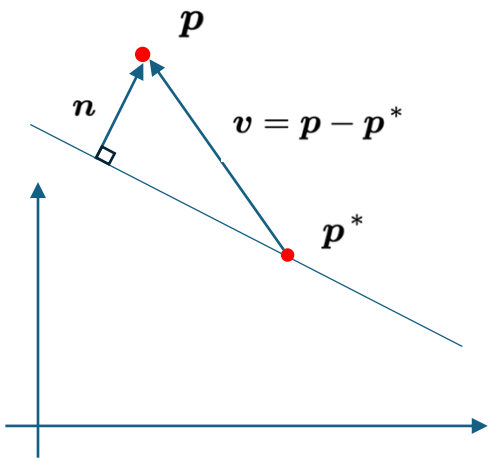
- Need a consistent measure of distance for comparison

$$d = \frac{\boldsymbol{n} \cdot (\boldsymbol{p} - \boldsymbol{p}^*)}{\|\boldsymbol{n}\|} = \frac{\boldsymbol{n} \cdot \boldsymbol{v}}{\sqrt{a^2 + b^2}}$$

$$\boldsymbol{n}^\top (\boldsymbol{p} - \boldsymbol{p}^*) = \boldsymbol{n}^\top \boldsymbol{p} - \boldsymbol{n}^\top \boldsymbol{p}^*$$

$$\boldsymbol{n}^\top \boldsymbol{p}^* + c = 0$$

$$\Rightarrow \quad \boldsymbol{n}^\top (\boldsymbol{p} - \boldsymbol{p}^*) = \boldsymbol{n}^\top \boldsymbol{p} + c$$



$$\boxed{d = \frac{\boldsymbol{n}^\top \boldsymbol{p} + c}{\sqrt{a^2 + b^2}}.}$$

Distance from a point to line

RANSAC - Inlier Test

Equivalently:

Scale ℓ so that $\sqrt{a^2+b^2}=1$:

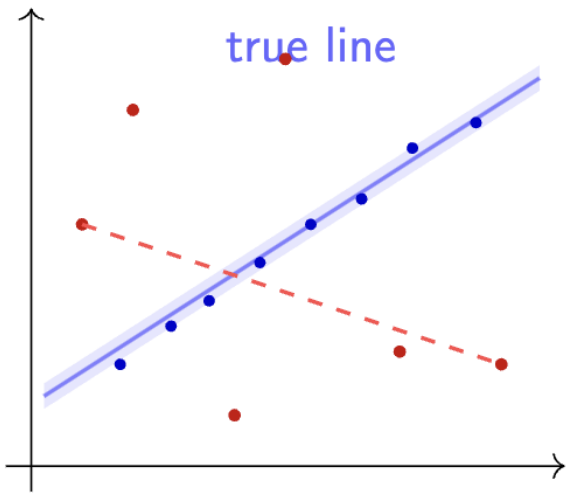
$$\ell \leftarrow \frac{1}{\sqrt{a^2+b^2}}\,\ell.$$

Then the signed distance in pixels is simply $r_i^{\text{dist}} = ax_i + by_i + c$.

Inlier test

$$\text{inlier if} \quad |r_i^{\text{dist}}| \leq \boxed{\tau.}$$ Parameter to tune

Example: noisy line



- Sample 2 points $\Rightarrow$ a candidate line
- Count inliers within distance τ
- Best hypothesis has the largest consensus

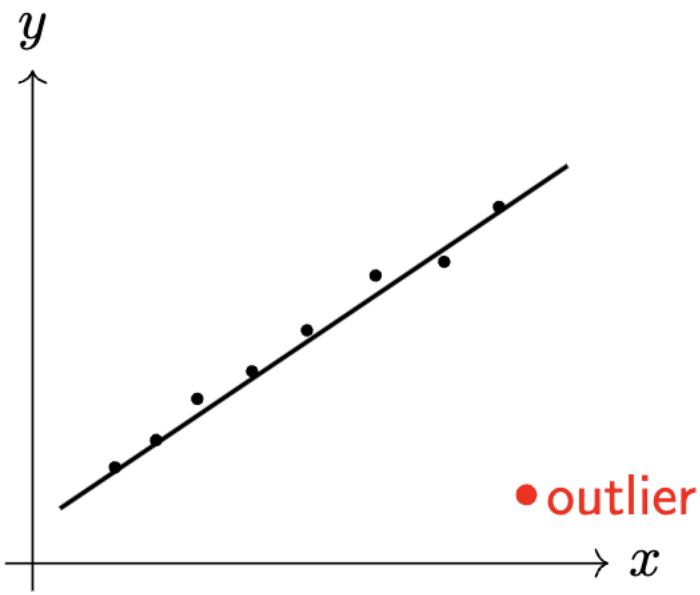
Example: noisy line

Geometric (orthogonal) distance

For line $\ell = (a, b, c)^\top$ and point $\boldsymbol{p} = (x, y)^\top$:

$$d(\boldsymbol{p}, \ell) = \frac{|ax + by + c|}{\sqrt{a^2 + b^2}}.$$

Least squares minimizes $\sum_i d(\boldsymbol{p}_i, \ell)^2$.



Full Algorithm

Algorithm

1. Initialize best model θ_{best} with score 0.
2. For $k = 1 \ldots N$:
   - Sample an MSS
   - Fit hypothesis θ_k
   - Compute inlier set $\mathcal{I}_k = \{i : r_i(\theta_k) \leq \tau\}$
   - If $|\mathcal{I}_k|$ better: refit on $\mathcal{I}_k$ and update best
3. Return θ_{best} and its inliers.

Can leave to the end to refit on all inliers

One trial

Let

- $\boxed{w = \text{inlier fraction (unknown)}}$ Your own guess
- $s =$ MSS size
- $p =$ desired success probability (e.g., 0.99)

If the inlier fraction is w, then

$$\Pr(\text{all } s \text{ are inliers}) = w \cdot w \cdots w = w^s.$$

So

$$\Pr(\text{trial fails}) = 1 - w^s.$$

N trials

Probability all N trials fail

$$\Pr(\text{fail in all } N) = (1 - w^s)^N.$$

We want

$$\Pr(\text{at least one success}) \geq p.$$

Using the complement event:

$$1 - (1 - w^s)^N \geq p \quad \Longleftrightarrow \quad (1 - w^s)^N \leq 1 - p.$$

Solve for N

$$1-(1-w^s)^N \geq p \quad \Longleftrightarrow \quad (1-w^s)^N \leq 1-p.$$

Taking $\log$ of both sides (note $0 < 1-w^s < 1$):

$$N \log(1-w^s) \leq \log(1-p) \quad \Longleftrightarrow \quad N \geq \frac{\log(1-p)}{\log(1-w^s)}.$$

N is the number of trials before reaching a probability of p

Refit on inliers

Once you have a consensus set $\mathcal{I}$, stop using random samples and solve a clean problem:

$$\theta^* = \arg\min_{\theta} \sum_{i \in \mathcal{I}} r_i(\theta)^2.$$

In case of a line: least square fitting on all inlier points

RANSAC for homography

Hypothesis

Sample 4 correspondences, solve for $\mathbf{H}$ (DLT), then score by reprojection error.

Residual

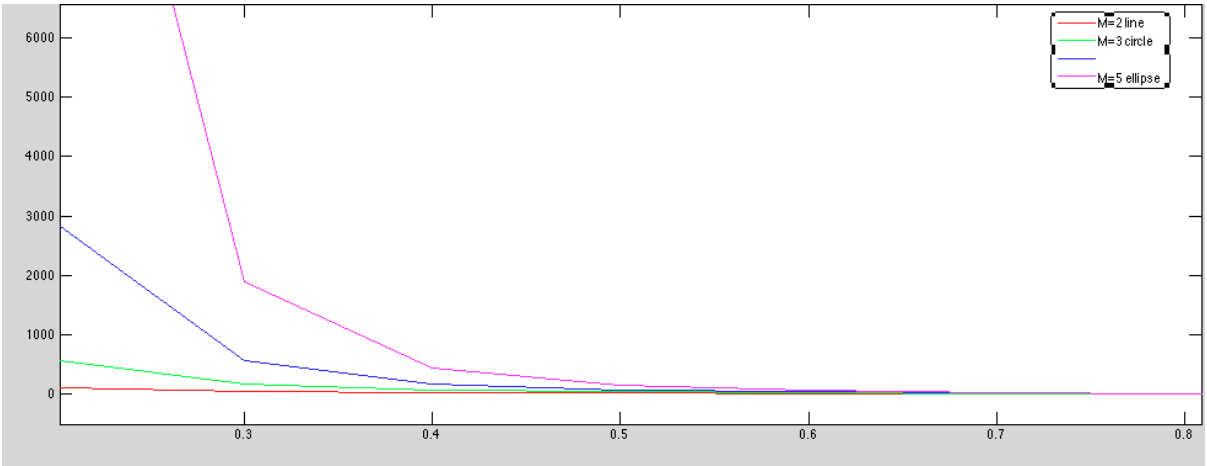
For correspondence $(\boldsymbol{X}_i \leftrightarrow \boldsymbol{x}_i)$ on the plane:

$$r_i(\mathbf{H}) = \|\pi(\mathbf{H}\boldsymbol{X}_i) - \boldsymbol{x}_i\|_2,$$

where $\pi([u\ v\ w]^\top) = (u/w,\ v/w)^\top$.

How tractable is RANSAC?

How tractable is RANSAC ?

$$k = \frac{\log(1-p)}{\log(1-\epsilon^M)}$$

Number of iterations needed to hit an inlier with probability 0.99 for the cases of line, circle, homograph, and ellipse fitting